1. $X = -123 = \overset{S}{1}\,\overset{64}{1}\,\overset{32}{1}\,\overset{16}{1}\;\overset{8}{1}\,\overset{4}{0}\,\overset{2}{1}\,\overset{1}{1}_{SM} = \overset{S}{1}\,000\;0101_{C_2} \longrightarrow$ CNT 7

$Y = -71 = 1100\;0111_{SM} = 1011\;1001_{C_2}$

Booth encoding

Q[0]	Q[-1]	Q
0	0	0
0	1	1
1	0	$\bar{1}$
1	1	0

$1011\;1001\;\overset{Q[-1]}{0}$

$Y^* = \bar{1}100\;\bar{1}011$ — operațiile care vor fi efectuate

CNT	A	Q (Y)	Q[-1]	M (X)
000	0000 0000	1011 1001	0	1000 0101
	⊕ 0111 1011 (−M)		⊖	−M: 0111 1011
	0111 1011 → RSHIFT			
	0011 1101	1101 1100	1 ⊕	
001	⊕ 1000 0101 (+M)			
	1100 0010 → RSHIFT			
	1110 0001	0110 1110	0 RSHIFT	
010	1111 0000	1011 0111	0 ⊖	
011	⊕ 0111 1011 (−M)			
	1 nu se interfereazā; 0110 1011 → RSHIFT			
	0011 0101	1101 1011	1 RSHIFT	
100	0001 1010	1110 1101	1 RSHIFT	
101	0000 1101	0111 0110	1 ⊕	
110	⊕ 1000 0101 (+M)			
	1001 0010 → RSHIFT			
	1100 1001	0011 1011	0 ⊖	
111	⊕ 0111 1011 (−M)			

① 0100 0100

nr întregi → shiftăm

0010 0010 0001 1101

1 nu face parte din nr

$P = (-123)\cdot(-71) = 8733 =$

S	8192	2048	512	256	128	64	32	16	8	4	2	1
0	0	1	0	0	0	1	0	0	0	1	1	1

16384 4096 1024